

Learning With Errors

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Outline

In this paper, I give an introduction to standard Learning with Errors (LWE), then ring based LWE. Then I provide a small visualization of standard LWE. Afterward, I briefly discuss some reasons why one might be used over the other and the hardness of the underlying LWE problem.

1 Standard LWE

Parameters

$$n \in \mathbb{Z}^+$$

$$q \text{ prime, } \mathbb{Z}_q$$

$$m \geq \lceil n \log(q) \rceil \quad [2]$$

$$A := \text{random } m \times n \text{ matrix} \quad (\text{column vectors form lattice})$$

$$a \in A \text{ is in } \mathbb{Z}_q^n \text{ (were not linearly independent in simulation)}$$

$$X \sim \mathcal{N}(\mu, \sigma^2) \quad (\mu = 0, \sigma = 1 \text{ seems to be effective})$$

Scheme

Bob wants to send a message $m \in \{0, 1\}$ to Alice.

Alice

Picks n .

Picks m, q, χ (error distribution).

Picks secret $s \in \mathbb{Z}_q^n$.

Computes random matrix A ($m \times n$).

Computes

$$b = As + e, \quad \text{where } (e \in \mathbb{Z}_q^m) \sim \chi$$

Publishes (A, b, q) as public key.

Once given (c_1, c_2) , computes

$$(c_2 - c_1 s) \pmod{q}$$

$$\equiv r^T b + m \left\lfloor \frac{q}{2} \right\rfloor - r^T A s$$

$$\equiv r^T (As + e) + m \left\lfloor \frac{q}{2} \right\rfloor - r^T A s$$

$$\equiv r^T e + m \left\lfloor \frac{q}{2} \right\rfloor =: v$$

If

$$v < \frac{q}{4} \quad \text{or} \quad v > \frac{3q}{4},$$

then message was 0 (Since, those values are closer to 0, than $\frac{q}{2}$ in modulo q).

Else, message was 1.

Bob

Uses Alice's public key (A, b, q) .

Picks random

$$r \in \mathbb{Z}_q^m$$

such that entries are in $\{0, 1\}$.

Computes ciphertext (c_1, c_2) where

$$c_1 = r^T A \pmod{q}$$

$$c_2 = \left(r^T b + m \left\lfloor \frac{q}{2} \right\rfloor \right) \pmod{q}$$

Sends (c_1, c_2) to Alice.

2 Ring-LWE

Defining Nega-cyclic Convolution

Pick integer $n = 2^k$, where integer $k > 0$.

Let q be a prime such that

$$q \equiv 1 \pmod{2n}.$$

Let ω be the n -th root of unity of $\mathbb{Z}_q$ (existence guaranteed from choice of q , as $n \mid q-1$ [8]).

Let ψ be the $2n$ -th root of unity of $\mathbb{Z}_q$ (existence guaranteed from choice of n, q , as $2n \mid q-1$ [8]).

Note that ψ is the $2n$ -th root of unity iff

$$\psi^2 \equiv \omega \pmod{q} \quad \text{and} \quad \psi^n \equiv -1 \pmod{q}. [8]$$

Denote the operation of nega-cyclic convolution by $\bar{\otimes}$.

Then define $\text{INTT}^{\psi^{-1}}, \text{NTT}^{\psi}, \bar{\otimes}$ as follows:

$$\bar{\otimes} : \mathbb{Z}_q^n \times \mathbb{Z}_q^n \longrightarrow \mathbb{Z}_q^n$$

via the mapping

$$(\alpha, \beta) \longmapsto \text{INTT}^{\psi^{-1}} \left(\text{NTT}^{\psi}(\alpha) \circ \text{NTT}^{\psi}(\beta) \right),$$

where $\circ$ is element-wise vector multiplication in $\mathbb{Z}_q^n$.

$$\text{NTT}^{\psi} : \mathbb{Z}_q^n \longrightarrow \mathbb{Z}_q^n$$

via the mapping

$$\alpha \longmapsto \hat{\alpha}, \quad \hat{\alpha}_j = \sum_{i=0}^{n-1} \psi^i \omega^{ij} \alpha_i \pmod{q}.$$

Then $\hat{\alpha}$ is said to be an NTT vector.

$$\text{INTT}^{\psi^{-1}} : \mathbb{Z}_q^n \longrightarrow \mathbb{Z}_q^n$$

via the mapping

$$\hat{\alpha} \longmapsto \alpha', \quad \alpha'_i = n^{-1} \sum_{j=0}^{n-1} \psi^{-i} \omega^{-ij} \hat{\alpha}_j \pmod{q}.$$

Then $\alpha = \alpha'$.

The caveat is that $\bar{\otimes}$ is still $O(n^2)$. However, we can reduce this to $O(n \log n)$ by replacing our defined NTT^{ψ} with the Cooley–Tukey (CT) algorithm for fast NTT, and replacing $\text{INTT}^{\psi^{-1}}$ with the Gentleman–Sande (GS) algorithm for fast INTT [8] (neither are discussed further, as it is beyond the scope of this paper).

The Scheme

Bob wants to send a message to Alice.

$R_q = \mathbb{Z}_q[x]/(x^n + 1)$ for $n = 2^k$ and $q \equiv 1 \pmod{2n}$.

Now

$R_q, + \leftrightarrow \mathbb{Z}_q^n, +$

$R_q, \cdot \leftrightarrow \mathbb{Z}_q^n, \bar{\otimes} [9]$

Alice

Computes $a \in R_q$, secret $s \in R_q$, puts coeff. of a, s into α, σ resp. Then she computes

$$\beta = \alpha \bar{\otimes} \sigma + \varepsilon \pmod{q}.$$

Publishes (α, β) as public key.

Bob

Chooses message $m \in R_q$, puts coeff. into $\mu \in \{0, 1\}^n$.

Uses Alice's public key (α, β) , chooses $t \in R_q$ with coeffs. in τ such that $\|\tau\|$ is small.

Computes new noise polynomials $e_1, e_2 \in R_q$ with coeffs. in $\varepsilon_1, \varepsilon_2$ resp. and creates ciphertext

$$(k_1, k_2) = \left(\alpha \bar{\otimes} \tau + \varepsilon_1, \beta \bar{\otimes} \tau + \varepsilon_2 + \mu \cdot \left\lfloor \frac{q}{2} \right\rfloor \right)$$

$$\approx (\alpha \bar{\otimes} \tau, \beta \bar{\otimes} \tau) = \left(\alpha \bar{\otimes} \tau, \alpha \bar{\otimes} \sigma \bar{\otimes} \tau + \mu \cdot \left\lfloor \frac{q}{2} \right\rfloor \right)$$

Sends (k_1, k_2) to Alice.

Once Alice has (k_1, k_2) , she computes

$$v = (k_2 - k_1 \bar{\otimes} \sigma) \pmod{q}$$

Then computes

$$\left\lfloor \frac{2v}{q} \right\rfloor \pmod{2} = \left\lfloor \frac{2(\mu \cdot \lfloor q/2 \rfloor \pmod{q})}{q} \right\rfloor \pmod{2}.$$

Since q is odd, set $q = 2\ell + 1$. Then

$$2 \left\lfloor \frac{q}{2} \right\rfloor \pmod{q} = 2 \left\lfloor \frac{2\ell + 1}{2} \right\rfloor \pmod{q}$$

$$= 2\lfloor \ell.5 \rfloor \pmod{q} = 2\ell \pmod{q} = q - 1.$$

That is, Alice has

$$\left\lfloor \frac{\mu \cdot (q - 1)}{q} \right\rfloor \pmod{2} = \mu$$

because (and let $\text{bit} \in \mu$)

If $\text{bit} = 0$, then

$$\left\lfloor \frac{0 \cdot (q - 1)}{q} \right\rfloor = 0 \equiv 0 \pmod{2}$$

If $\text{bit} = 1$, then

$$\left\lfloor \frac{1 \cdot (q - 1)}{q} \right\rfloor = 1 \equiv 1 \pmod{2}$$

Thus Alice has successfully recovered the original message from Bob.

Differences

Some notable advantages of ring based LWE over standard LWE is that the public key is smaller and that the computation can be more efficient. However the cost of this is the complexity of the scheme, which makes standard LWE easier but less efficient to use.

Hardness

The underlying problem of both ring based LWE and standard LWE is based on computing a small enough error in order to find a sufficient approximation for the secret key to calculate b (for standard LWE) and β (for ring based LWE).

This involves trying to find a unique closest lattice point to a given point. That is, in the average case, LWE is as hard as the Bounded Distance Decoding problem where the bound is usually the norm of the error vector.

However, any error, which is used to approximate the secret, that is smaller than the original error should suffice. That is to say, the smaller the error used for the public key, the more secure the encryption.

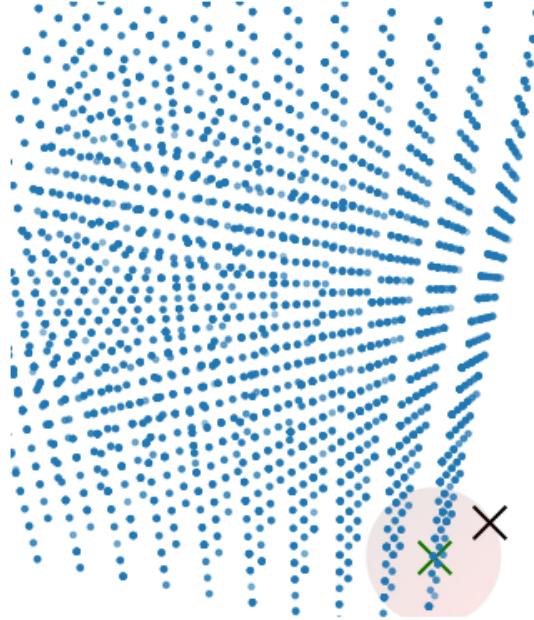


Figure 1: example simulation of standard LWE with $n=3$, $q=11$, $m=\lceil n * \log(q) \rceil$, error is represented as the radius of sphere. The green X is the point $A*s \bmod q$ (a guaranteed lattice point) and the black X is the point $b = A*s + e \bmod q$.

Notice in figure 1, we used the true secret key. In practice however, this is not known and the attacker would have to solve for a sufficient error vector, e' , which corresponds to an approximation of s , s' where $A*s' \bmod q$ is a lattice point. inside this sphere.

By doing this, the attacker can use s' to compute $A*s' + e' \bmod q \approx b$, hence being able to decrypt the intercepted ciphertext with high probability.

References

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